Take some shapes like this:

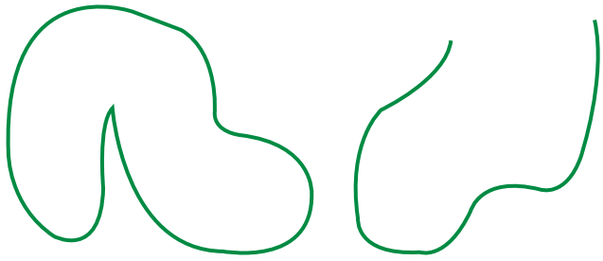


Fig. 4.1

In both of them, there is a *curve* forming the shape: one is a closed curve, enclosing a region, and the other is an open curve. We can use a rope (or a thick string) to measure the **length** of the open curve and the length of the boundary of the region in the case of the closed curve.

Curves are tricky, aren't they? It is so much easier to measure length of straight lines using the scale, isn't it? Consider the two shapes below.

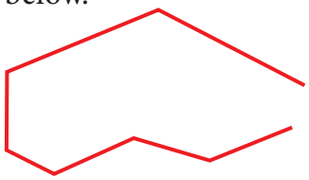


Fig. 4.2

Now we are going to focus our attention only on shapes made up of straight lines and on closed figures. As you will see, there is plenty of interesting things to do. Fig.4.2 shows an open figure.

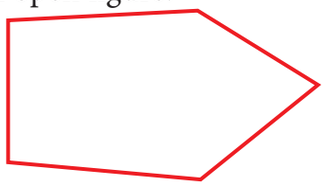
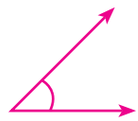


Fig. 4.3

We not only want to draw such shapes, we want to compare them, measure them and do much more. For doing so, we want to **describe** them. How would you describe these closed shapes? (See Fig 4.3) They are all made up of straight lines and are closed.

## 4.2 Types of Angles-Recall

Plumbers measure the angle between connecting pipes to make a good fitting. Wood workers adjust their saw blades to cut wood at the correct angle. Air Traffic Controllers (ATC) use angles to direct planes. Carom and billiards players must know their angles to plan their shots. An angle is formed by two rays that share a common end point provided that the two rays are non-collinear.



**acute**

an angle that is less than 90°



Fig. 4.4



**right**

an angle that is exactly 90°

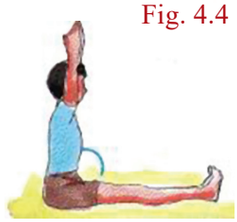
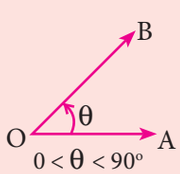
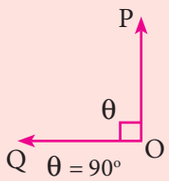
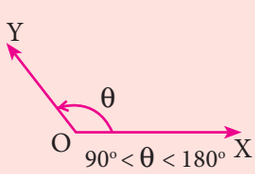
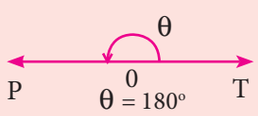
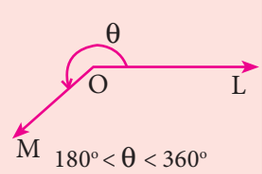


Fig. 4.5

| Acute Angle | Right Angle | Obtuse Angle | Straight Angle | Reflex Angle |
|---|---|---|---|---|
|  |  |  |  |  |
| $0 < \theta < 90^\circ$ | $\theta = 90^\circ$ | $90^\circ < \theta < 180^\circ$ | $\theta = 180^\circ$ | $180^\circ < \theta < 360^\circ$ |



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